



devices to get a single link to the Internet with increased bandwidth.

Before you start off...

This method of using two different modems and two phone lines to increase the overall bandwidth is called Remote Access Service (RAS). The process involves attaching two internal or external modems to a single system. One point that needs to be kept in mind is that your ISP should also provide this option with their services.

The best part is that operating systems from Windows 98 onwards come with this feature built into their Dial-up Networking. For older operating systems such as Windows 95, you would need to download the latest Dial-up networking patch in order to enable the use of Multilink PPP features. Using Multilink PPP, data packets from your PC are broken up into parts and each of these parts is sent from your two modems onto the ISP's modems where they are reassembled, thus providing twice the bandwidth.

One of the best and easiest ways to get Multilink PPP working is to use Windows 98. You can use two simple dial-up 56 Kbps modems, for example, and gain an overall speed of 112Kbps (56 Kbps x 2). Certain free software, such as MultiRAS allow you to use this feature even if you don't have the latest Windows Dial-up networking update on your system.

Lets get to it...

Installing the 2 modems: First of all, if you're installing internal modems, free up any PCI or ISA slots that you may be using. In case, you don't have any free PCI or ISA slots on your system, you will need to remove one of the cards. Internal modems can be used in conjunction with external modems. For installing an external modem, connect the modem to the external COM port or USB port. Connect each of the modems to the two phone lines and also the power supply if needed, in the case of external modems.

Drivers for modems: Boot your machine and wait for the two modems to be auto detected. In case the modems are not detected, you can go to **Start > Control Panel** and then to the Modem options. You can then choose to auto detect the modem by selecting it or you can manually set up each modem by choosing the correct drivers and specifying the serial ports to which they are connected. You can also go to the **Control Panel > Add New Hardware** and do the same. Add the necessary network protocols such as TCP/IP and IPX/SPX, if needed.

Creating a connection: Next, you need to set up a dial-up connection. To do this, go to Dial-up Networking and click on Create New Connection. Enter the name for the ISP and other information such as the phone number to be dialled to connect to the ISP.

Setting up the Multilink protocol: Once a connection has been created using Dial-up Networking, right-click on the ISP's icon and choose Properties. Now click on the Multilink tab. You can now add, remove or edit various devices.

Dialling: The final step is to dial to the Internet using your dual modem connection. Go to Dial-up Networking and double-click on your ISP icon. Enter your username and password and start dialling. After you're online, you'll have twice the bandwidth as compared to using a single Internet access method.

No support for Multilink PPP?

It could also happen that your ISP may not support the Multilink PPP protocol, thus bringing all your plans for extra bandwidth to a halt. However, certain hardware and software are available that can help you overcome this obstacle. For example, a software called SurfDoubler allows you to use two different ISPs and two different Net accounts to double your bandwidth via dual modems. This software downloads different pages through each of the ISPs that you are connected to using the dual modems. The software also includes other features such as security and firewall functionality, built into it. Midcore is a similar software.

Another very popular solution in Internet bandwidth bonding lies in the use of special hardware such as the WebRamp device. WebRamp is like a gateway to the Internet for a network and is capable of being connected to more than one modem or ISDN terminal adapter so as to bond the bandwidth of these devices.

Advantages of dual modems

While a dual modem connection does mean double bandwidth, you will also be paying double your existing phone and Internet access bills as your ISP will charge you more for the two connections. However, the resultant bandwidth improvement is very apparent and this is also a very cost-effective solution in terms of initial investment. A dual 56 Kbps modem would result in download speeds of around 10 Kbps, which is double the bandwidth that you would normally get on a single modem.

An office user can start a proxy server on a particular machine that uses these two modems to provide Internet access. This is useful even for home users who don't have broadband. With the capability of even bonding ISDN and cable Internet connections, the twofold increase in bandwidth could open a range of broadband possibilities.

An especially useful application of this technology would be in cyber cafes, where you could merge the bandwidth of various devices for providing Internet connectivity to a group of users. 

Glossary

RAS: Remote Access Services is a function that allows users to connect to a network or server using a modem.

TCP: Transmission Control Protocol, one of the most widely used protocols on networks and on the Internet, to allow computers to communicate with each other.

ISDN: An acronym for Integrated Services Digital Network, which is a communication standard for computers and gives higher bandwidth as compared to normal phone lines. It also allows simultaneous use of telephony services while using the Internet.

PPP: Point to Point Protocol is used to connect computers to each other using modems or other such communication devices.

DSL: Digital Subscriber Line is a major broadband connection available for office and home users. Variants of DSL are Assymetric Digital Subscriber Lines (ADSL) and High data-rate DSL (HDSL).